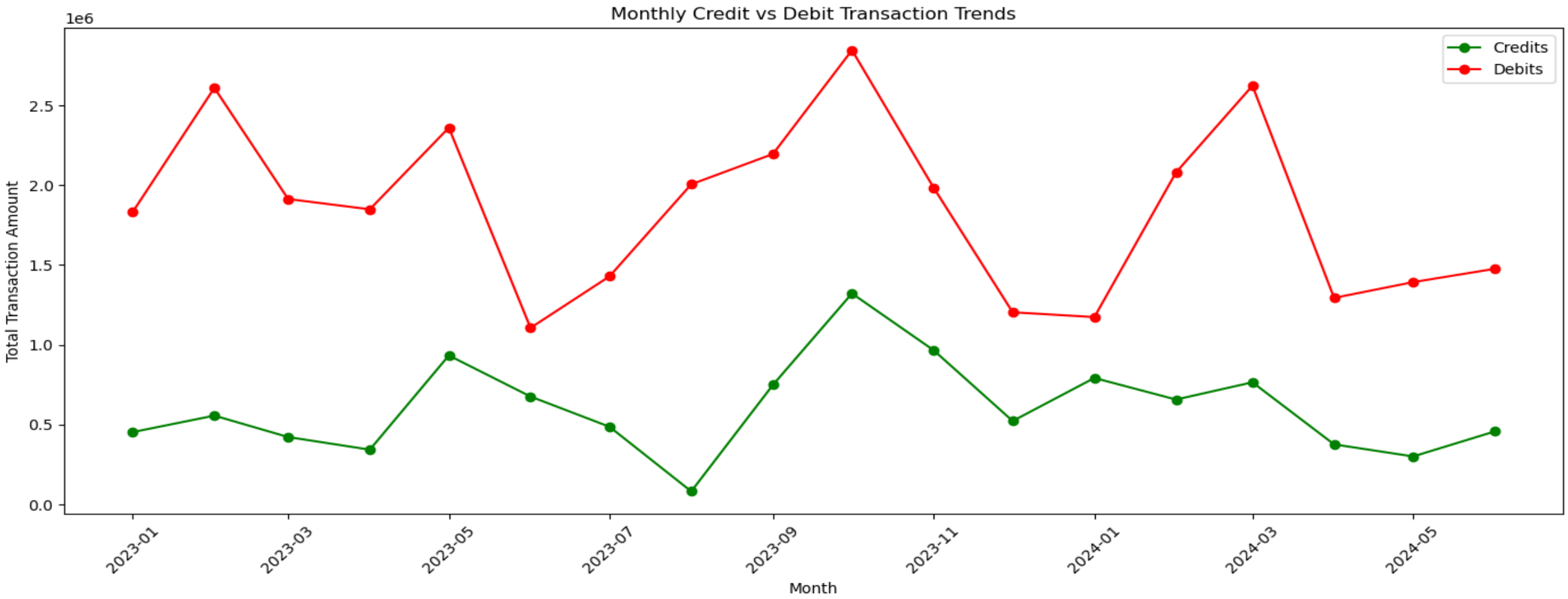
Financial Risk Analysis with Python-Blackrock

Task 1: Data Cleaning and Formatting

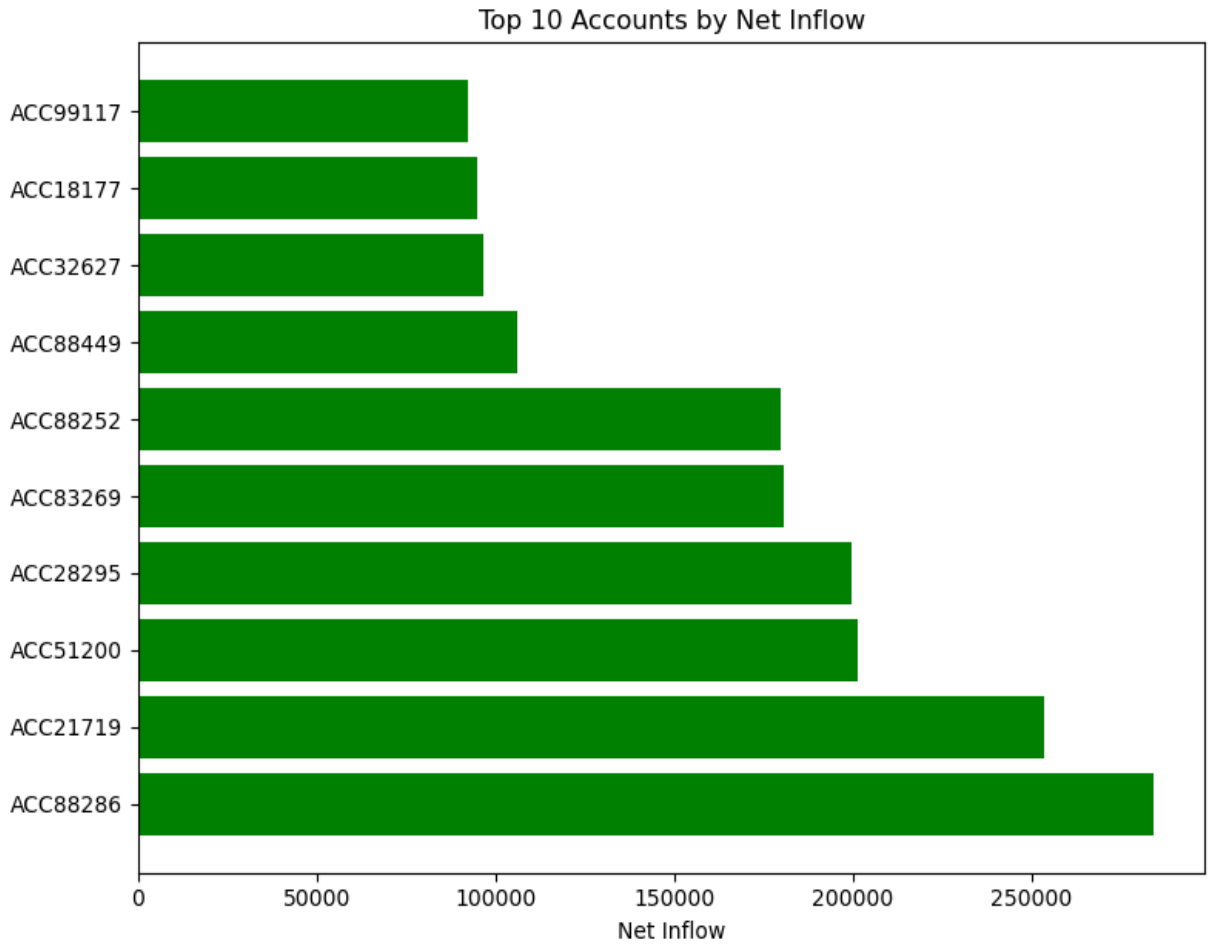
- First of all, the data in CSV format was loaded in the Jupyter notebook.
- After importing the libraries such as pandas and numpy, and implementation of the algorithms, we found that there are 0 duplicate, missing and special values in the columns.
- All the currency amount column are already in the numeric format, that requires no further conversion.
- The date columns have been formatted into YYYY-MM-DD Format with datetime64[ns] dtype.
- The account types and transaction categories were already clean and standardized in the format.

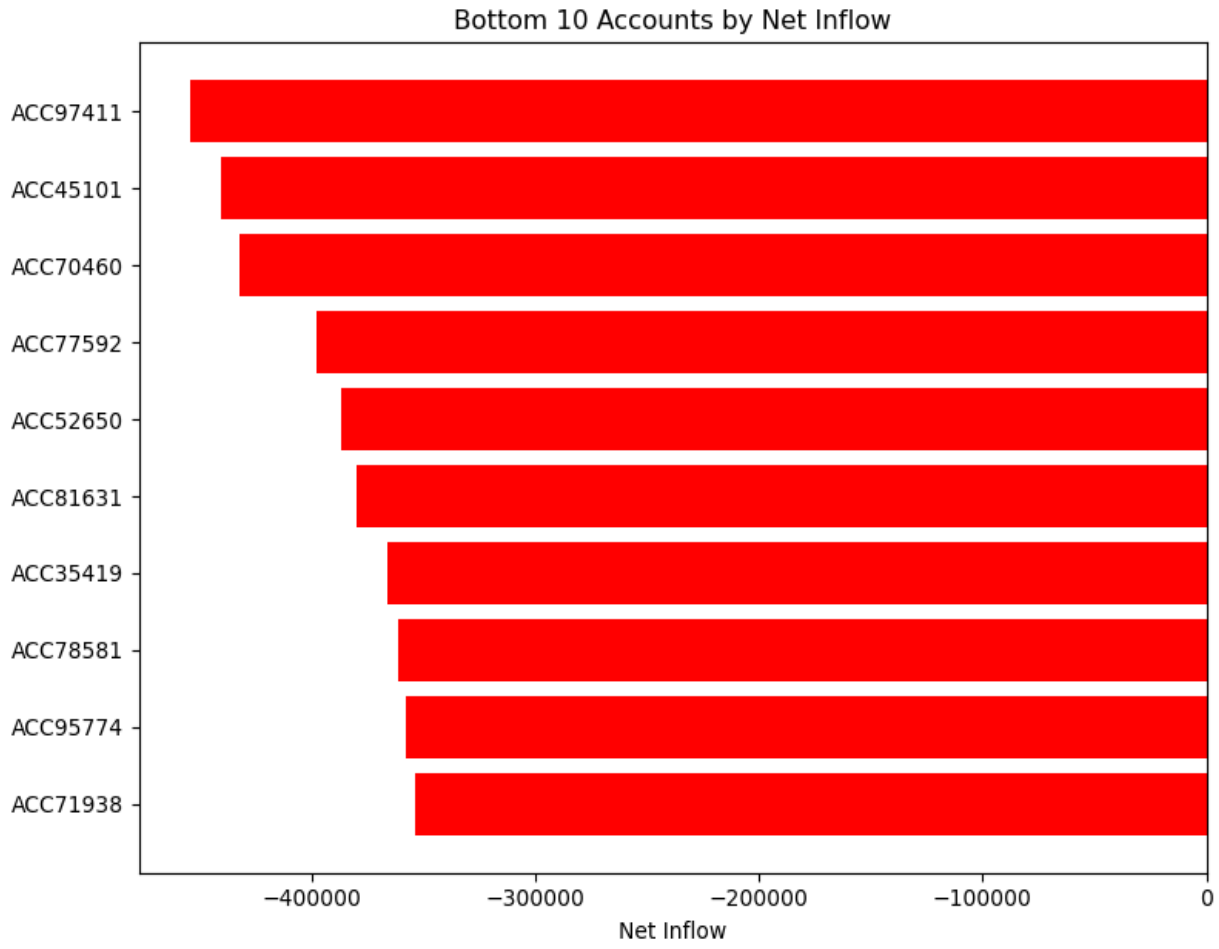
Task 2: Descriptive Transactional Analysis

- The total credit transactions is 182 and the total debit transaction is 618.
- Trends in total credit vs debit over time is plotted below:



- The top performing account is ACC88286 with net inflow +₹2,84,230.60.
- The bottom performing account is ACC97411 with net inflow -₹4,54,306.22

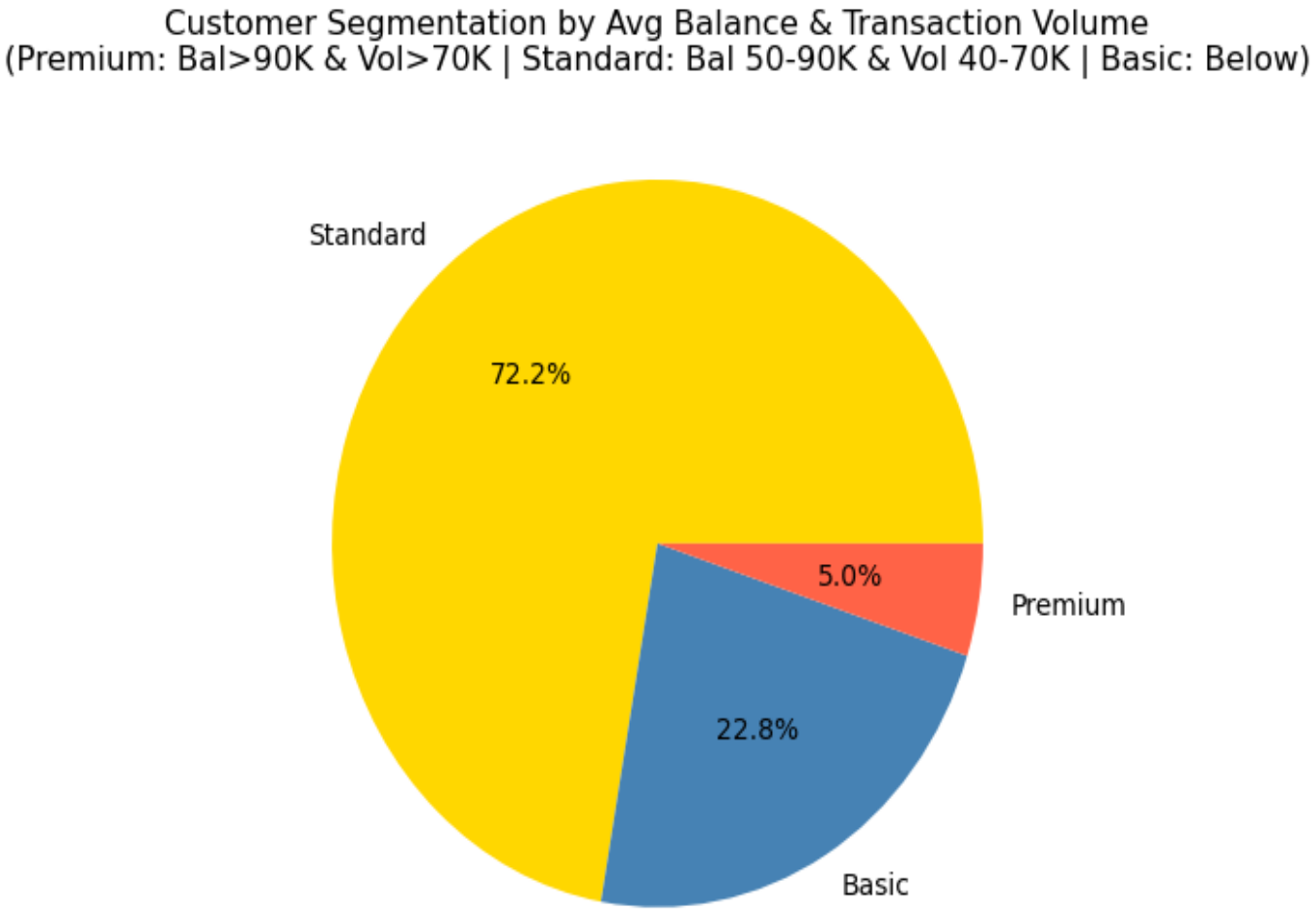




- There are 172 accounts (89.6%) flagged dormant with 60+ day transaction gaps.

Task 3: Customer Profile Building

- Upon the activity level grouping, there are 105 medium, 45 high and 42 low activity accounts.
- The Average balance across all customers = **₹72,602** and avg transaction volume = **₹55,755.**



The profile for High Net Inflow Accounts	
Metric	Detail
Threshold (90th percentile)	₹55,893.97
Total Accounts	20
Top Account	ACC88286 with ₹2,84,230.60 net inflow
Dominant Account Type	Credit accounts dominate this profile
Dominant Region	North & West regions lead

- **The profile for High Frequency Low Balance Accounts:**

Metric	Detail
Frequency Threshold (75th percentile)	5+ transactions
Balance Threshold (25th percentile)	Below ₹63,455
Total Accounts	15
Top Account	ACC41829 with 13 transactions
Risk Indicator	High activity but low balance — potential liquidity risk

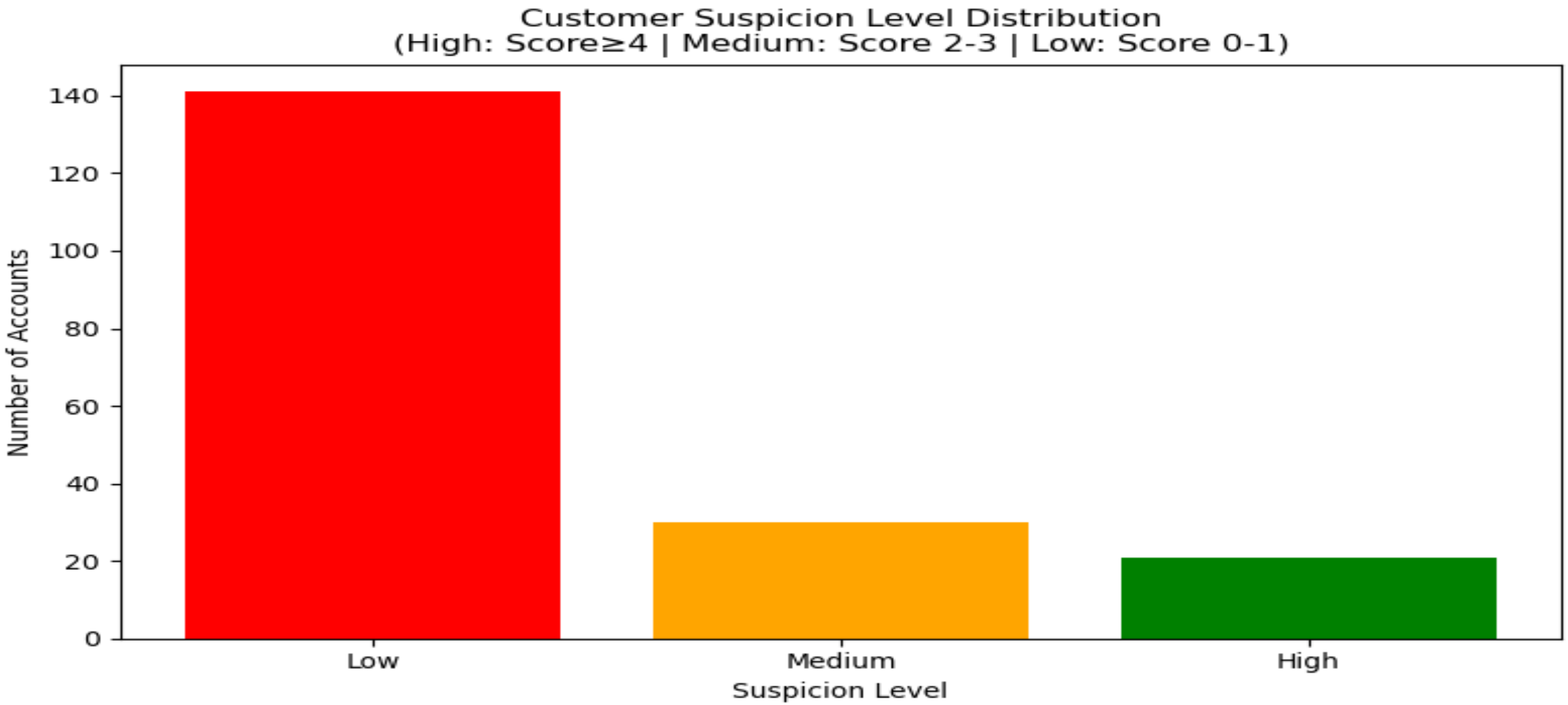
- **The profile for Negative/Near-Zero Balance Accounts:**

Metric	Detail
Threshold	Below ₹1,000
Total Accounts	0
Reason	All accounts maintain average balance above ₹1,000 — overdraft instances are isolated transactions, not sustained low-balance patterns

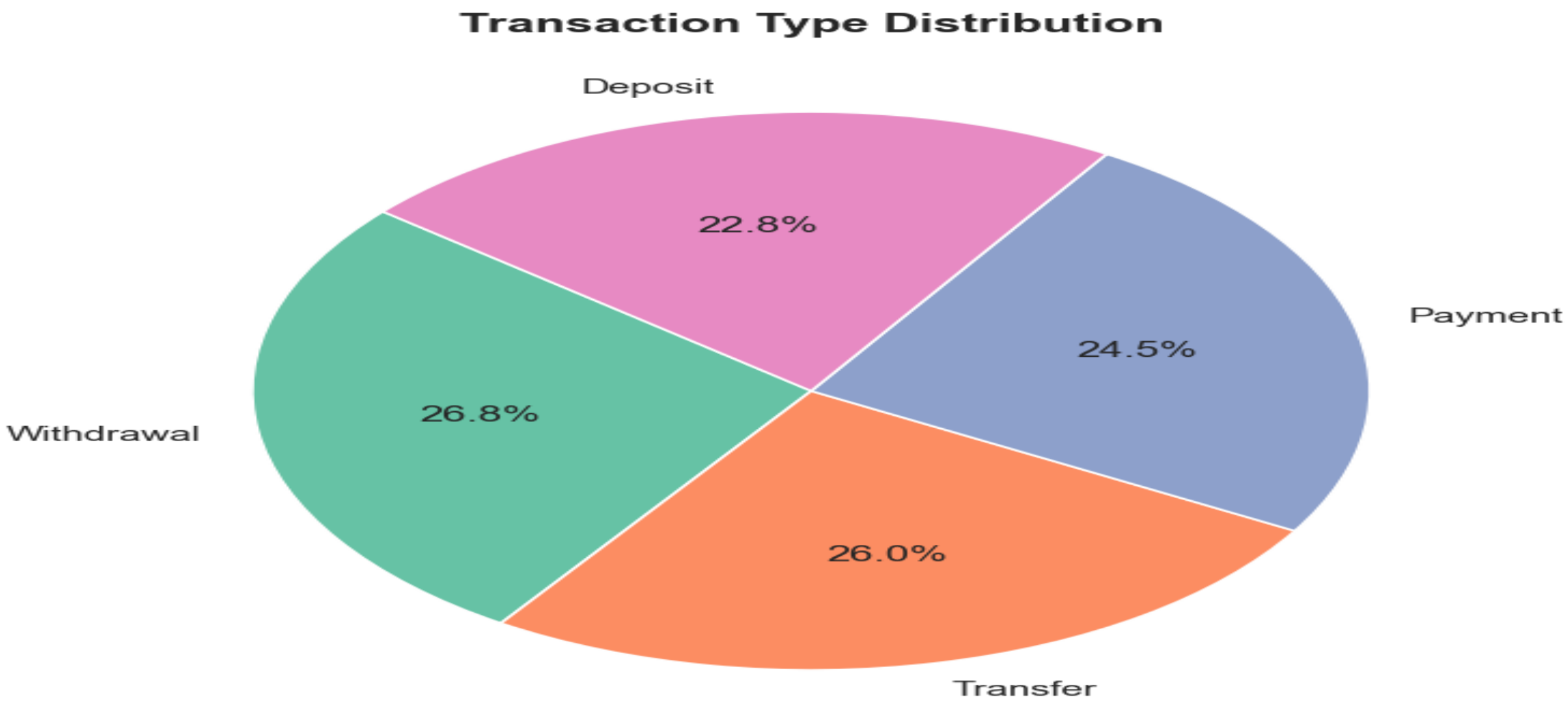
Task 4: Financial Risk Identification

- There are 10 accounts that are frequent large withdrawals with 3+ large withdrawals threshold.
- **ACC77638** is the most aggressive — 4 large withdrawals flagged.
- There are 46 accounts with CV > 56.77%.
- **ACC78589** is the most volatile with CV of **252.16%**
- There are 5 anomalies using IQR method and 4 anomalies using Z-Score method.
- **ACC45101 (CUST1888)** is the most extreme anomaly at ₹1,65,00 flagged by both methods.
- **ACC83848** was caught only by IQR .
- There are 21 accounts with score >= 4, having high suspicious level.
- **ACC45101** is the most suspicious account with a perfect score of **8/8** , flagged on all 5 risk indicators

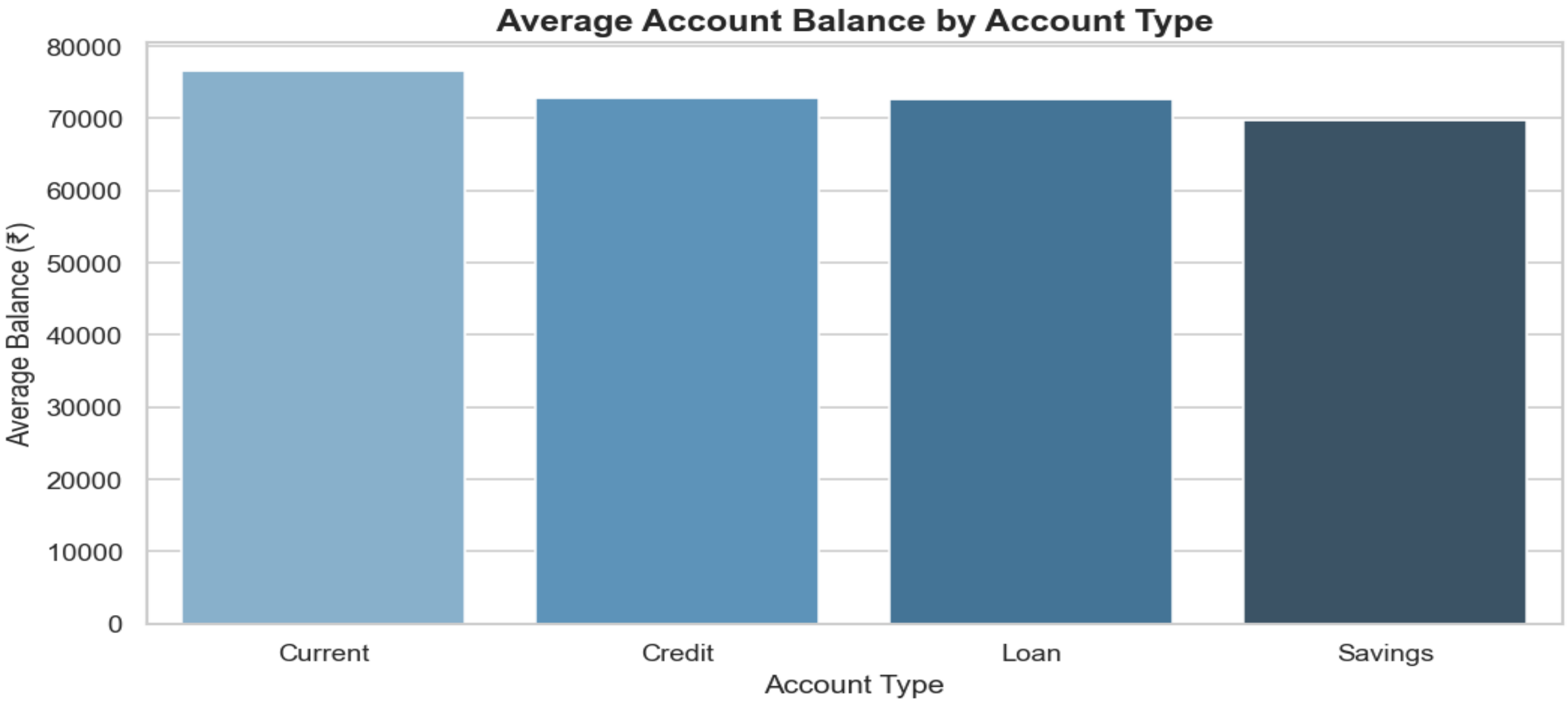
Pictorial representation of above findings.



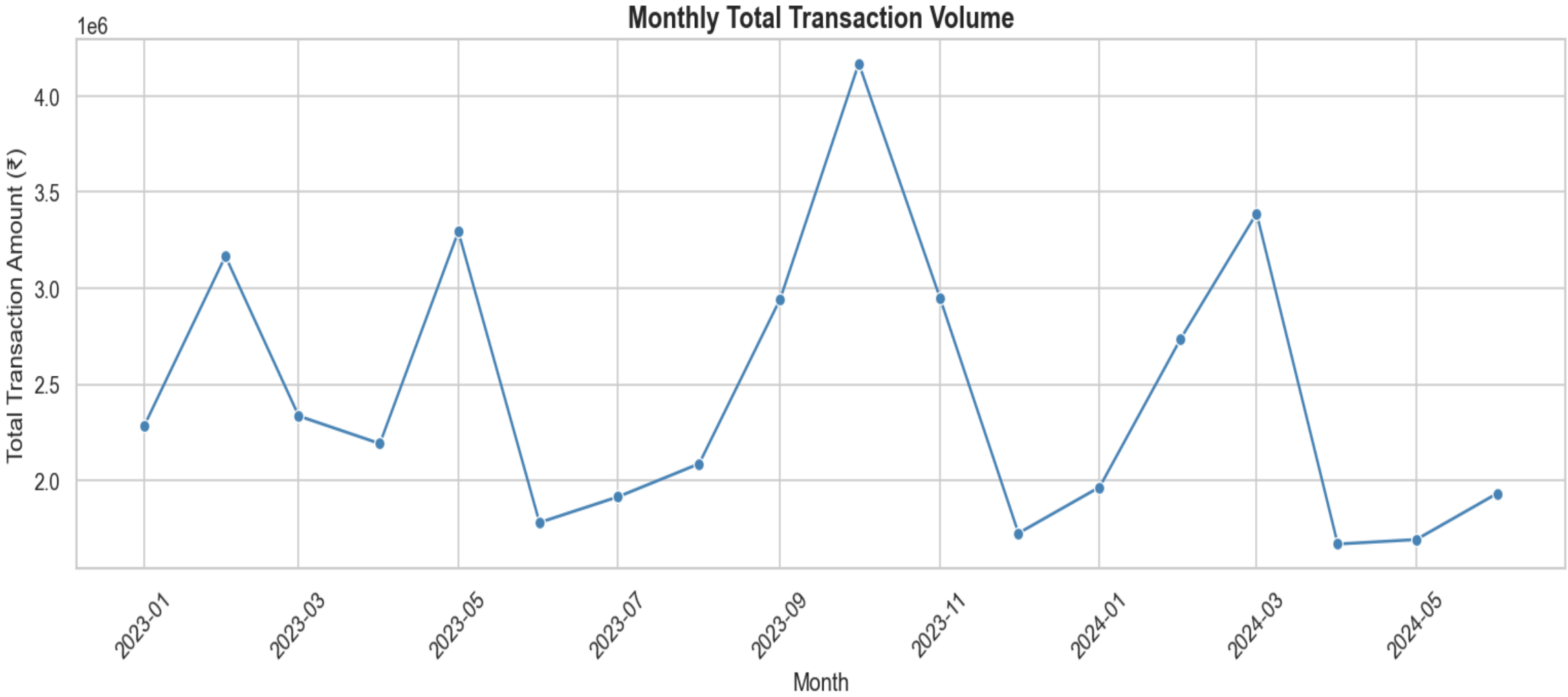
Task 5: Visualisation

- 

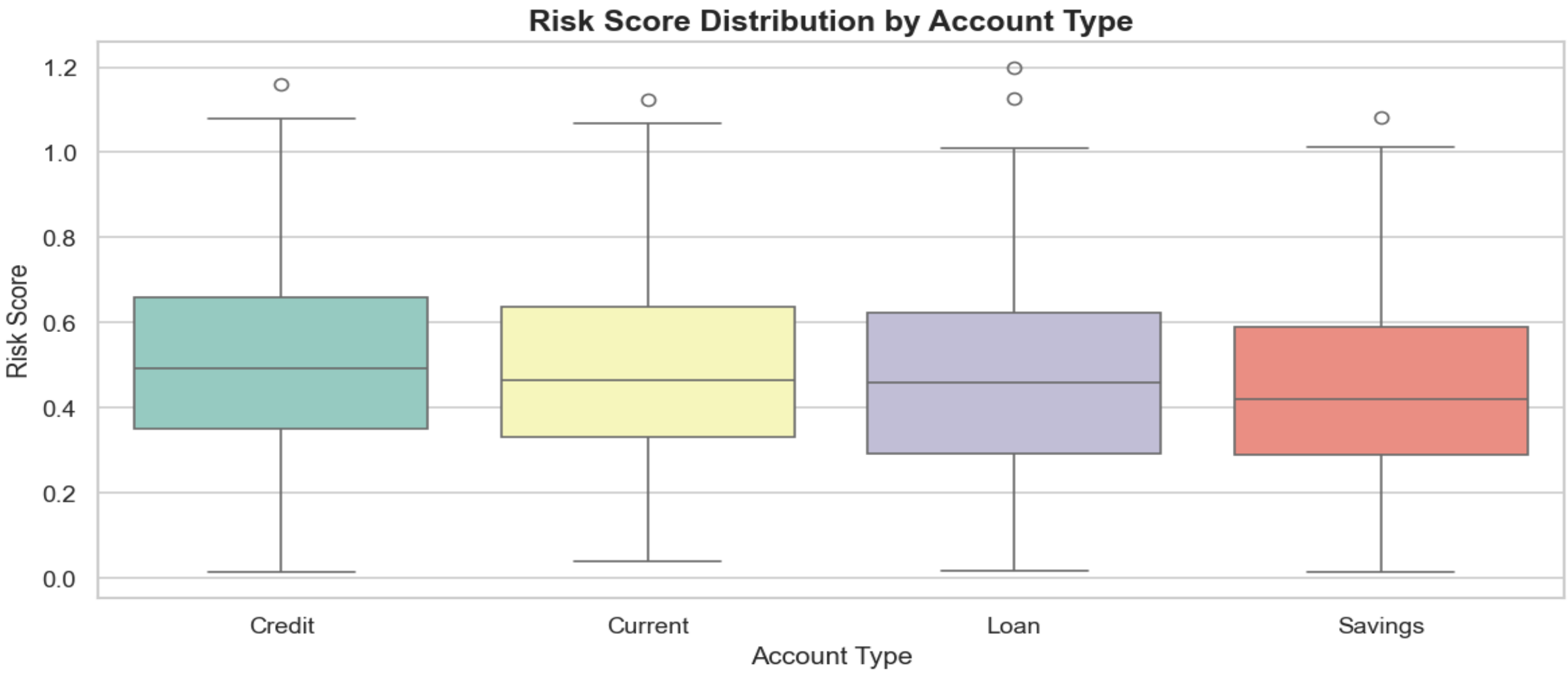
Average Account Balance by Account Type



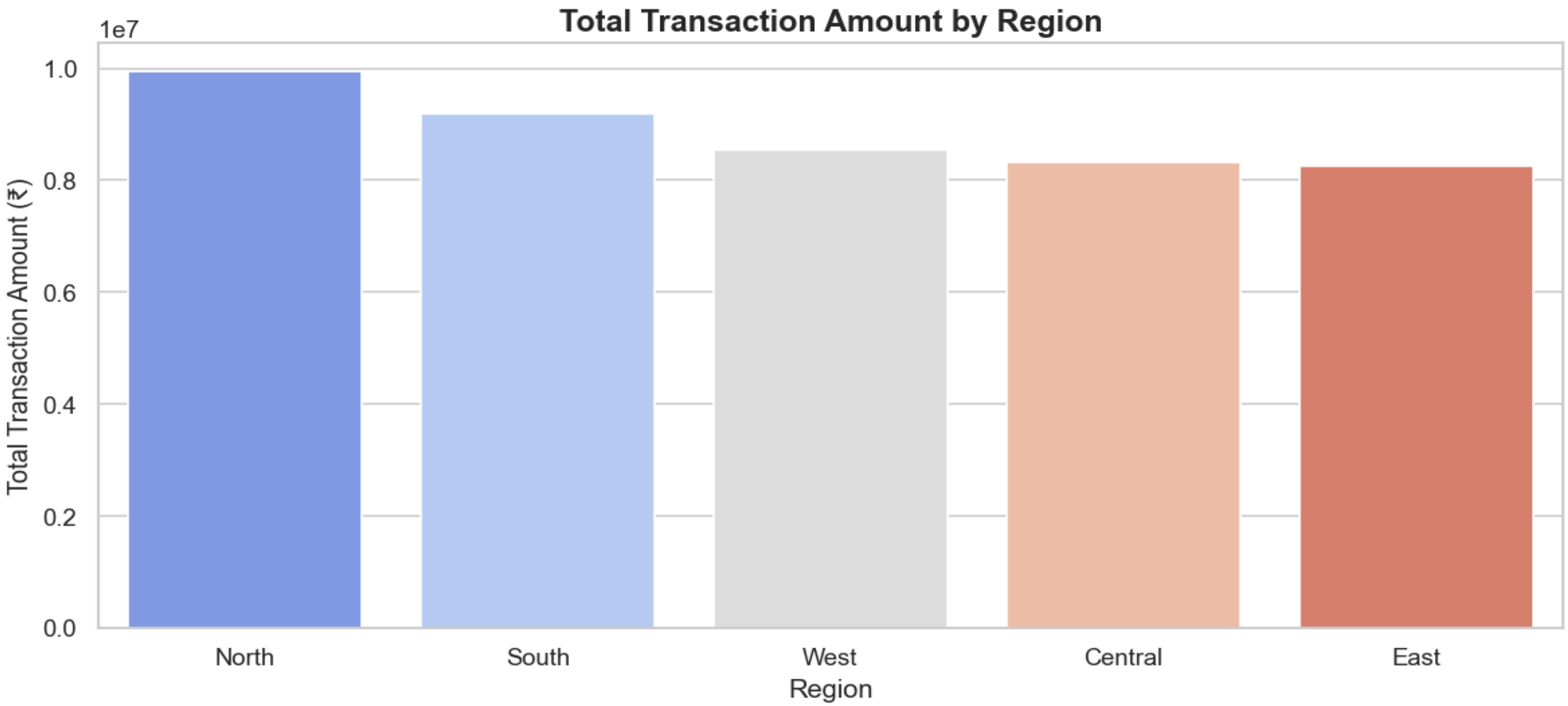
Monthly Total Transaction Volume



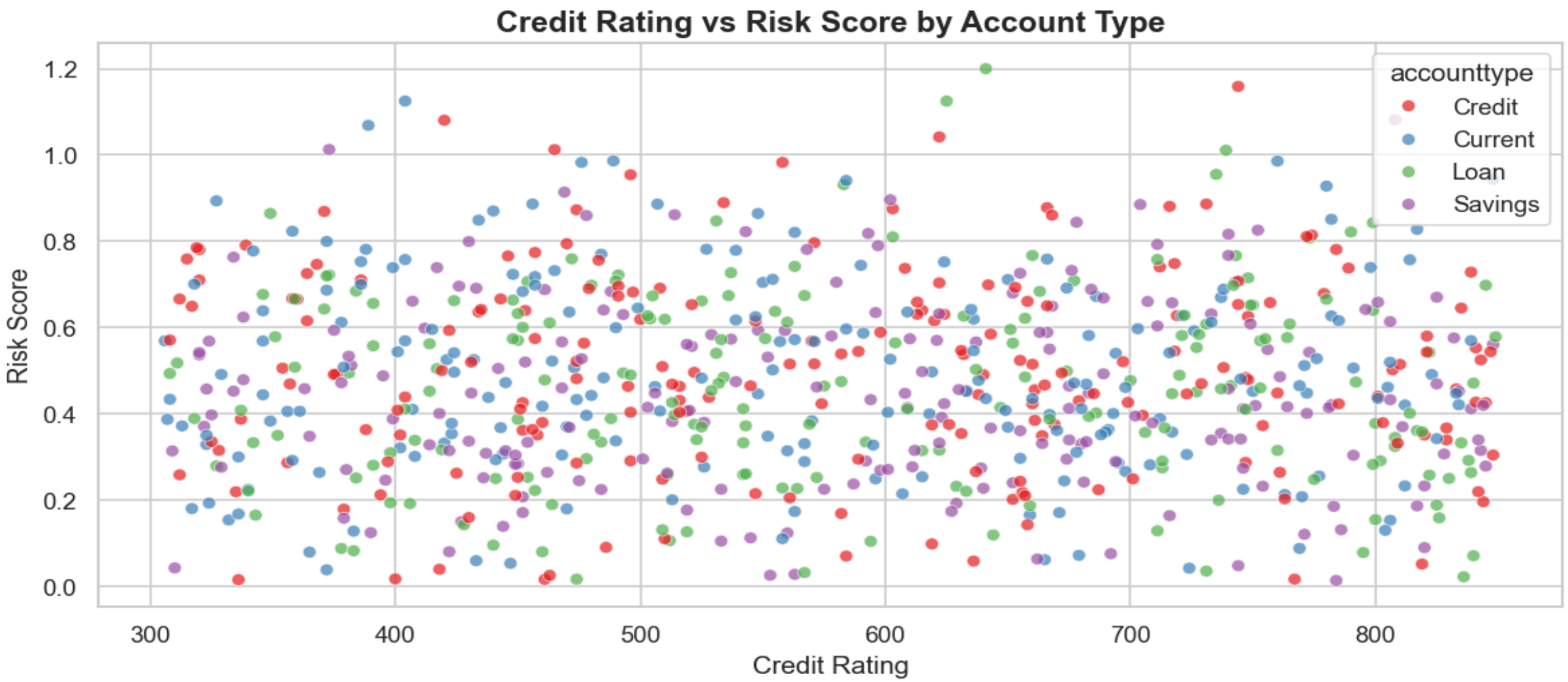
Risk Score Distribution by Account Type



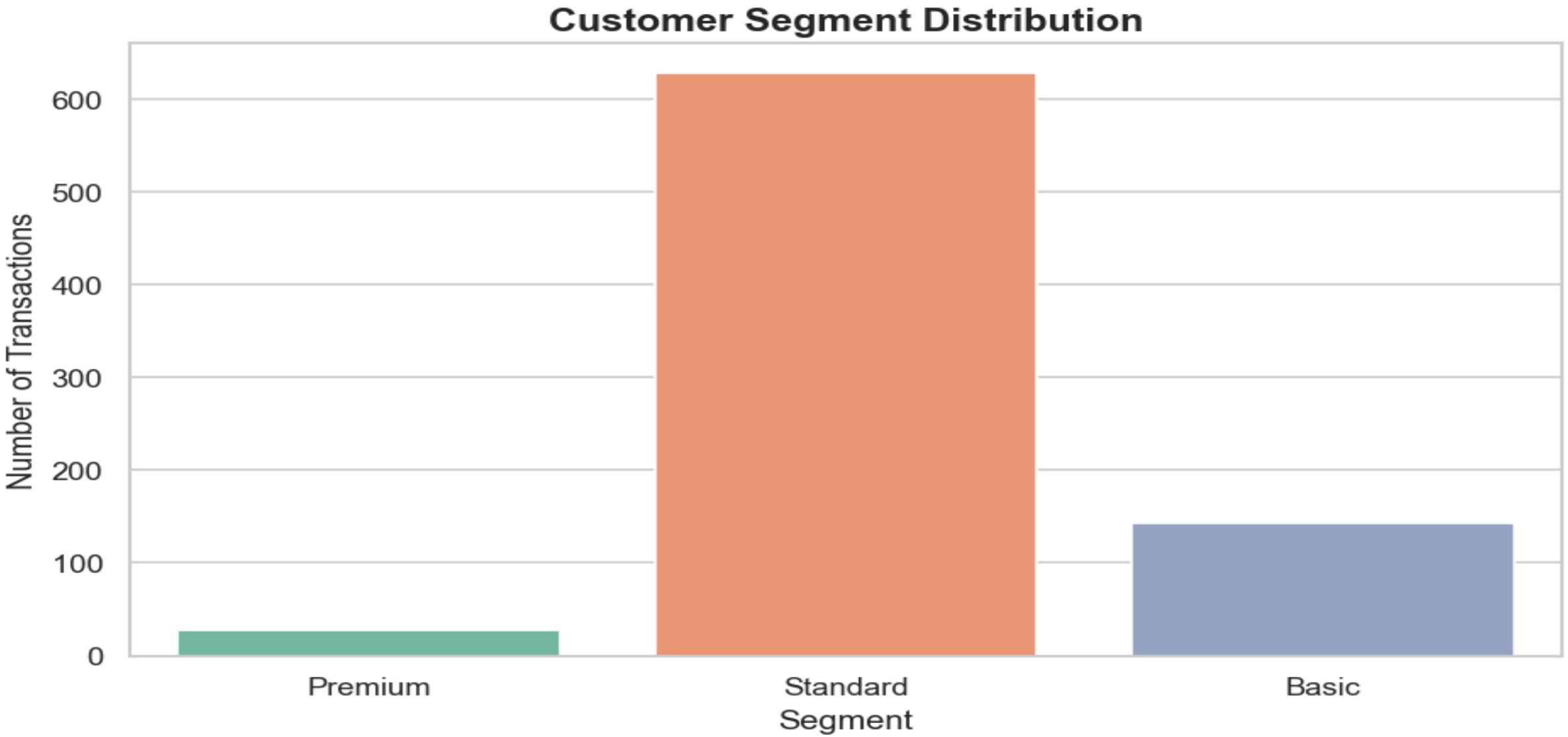
Total Transaction Amount by Region



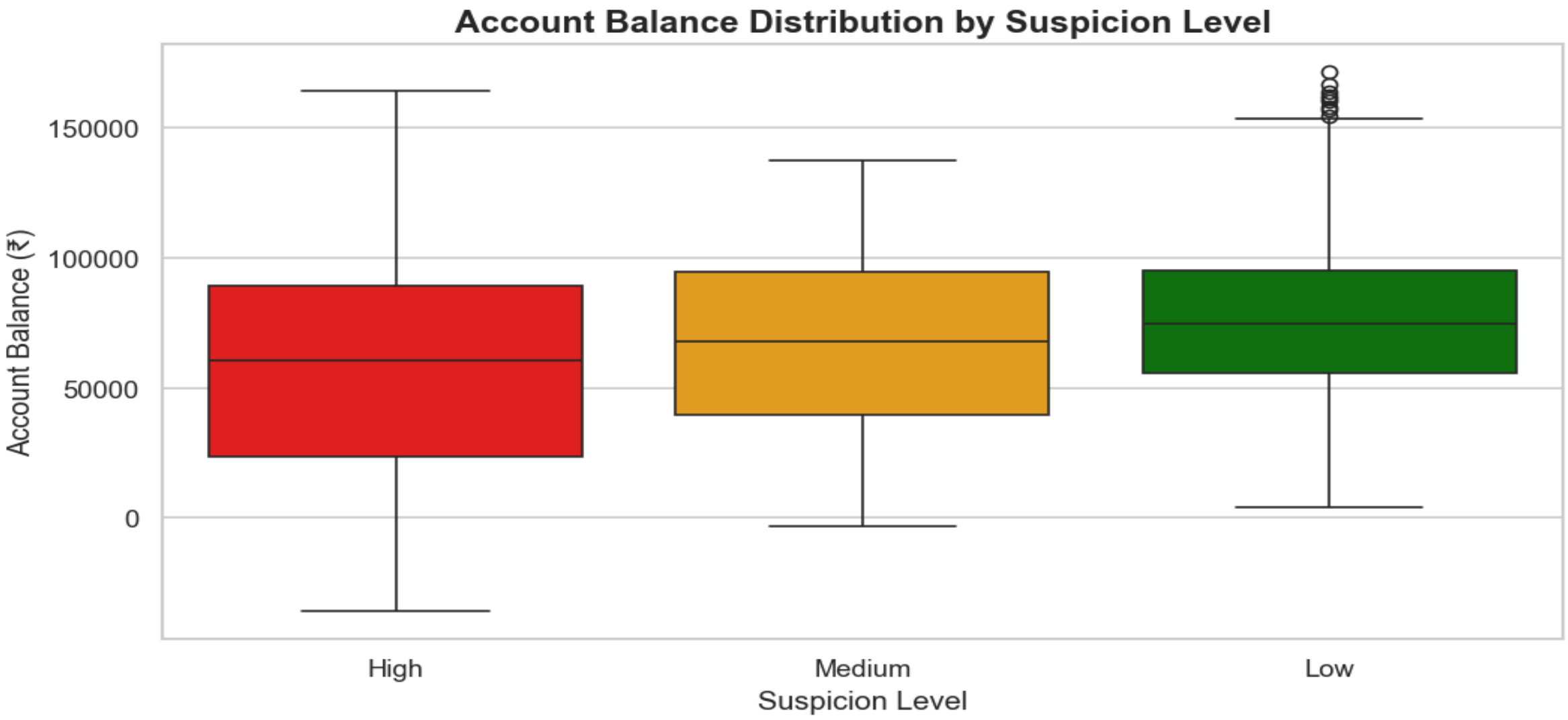
Credit Rating vs Risk Score by Account Type



Customer Segment Distribution

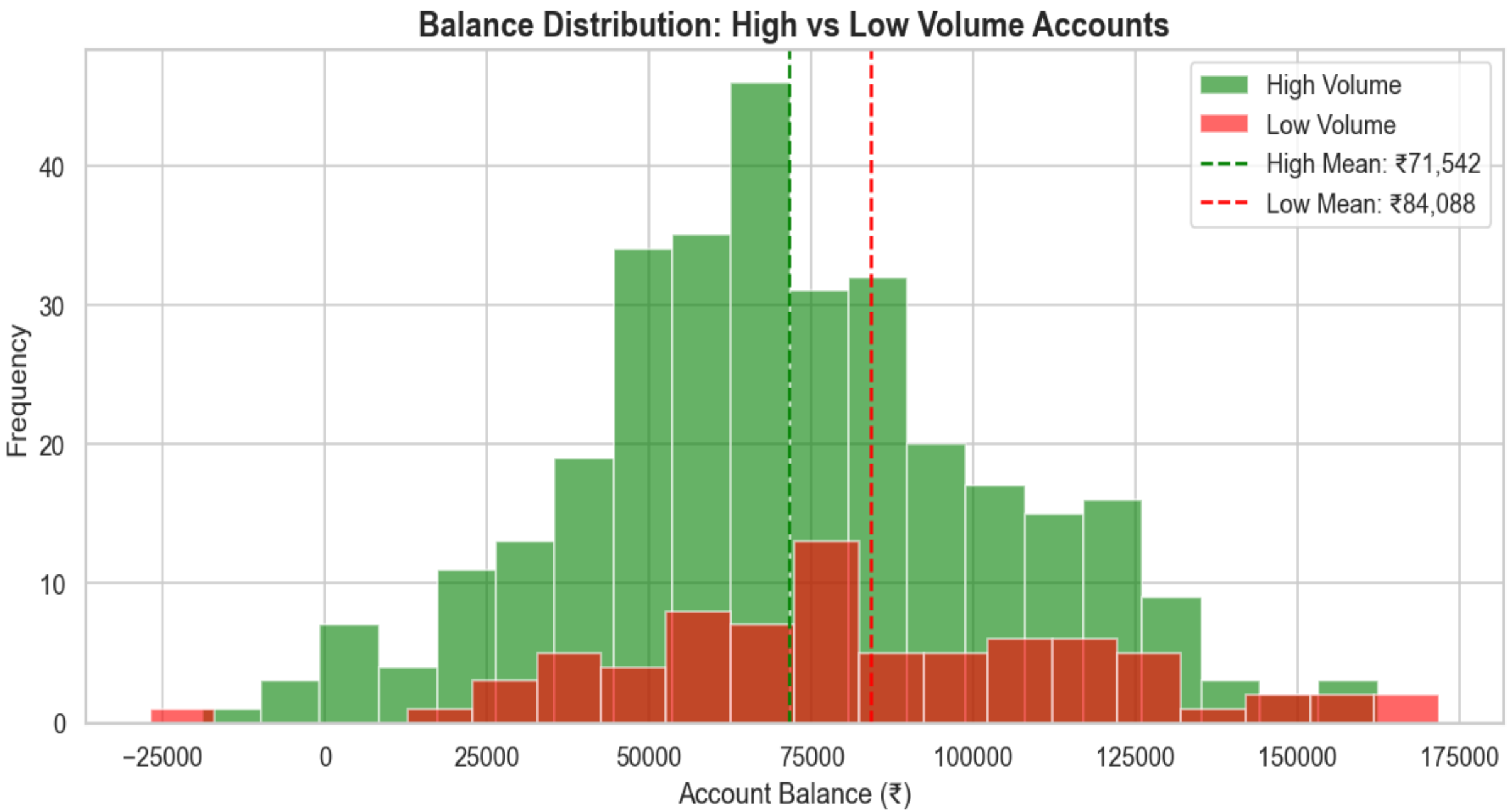


Account Balance Distribution by Suspicion Level



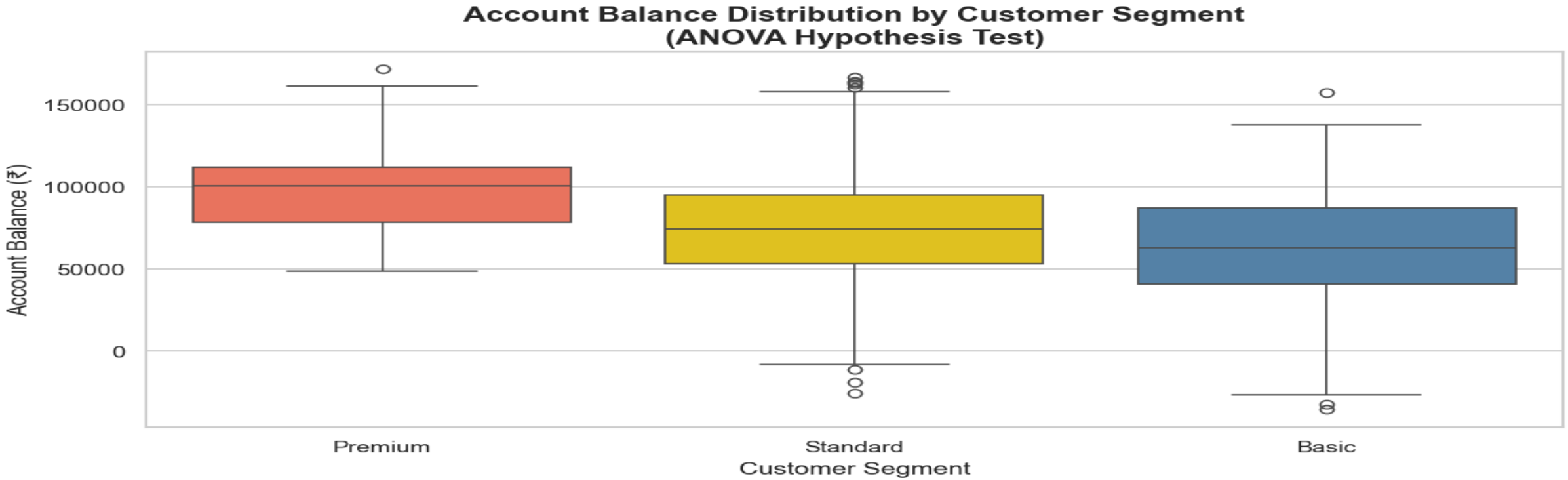
Task 6: Hypothesis Testing

- There are 321 transactions in high volume accounts and 76 transactions in low volume accounts.
- The mean balance for low volume accounts is ₹84,088.33 while the mean balance for high volume accounts is ₹71,541.72
- Surprisingly **Low volume accounts have HIGHER average balance.**
- T-statistic is **negative** — confirming high volume accounts actually have lower balances.
- **High transaction frequency does NOT lead to higher balances.**
- Below is the bar graph representation of Balance Distribution stating High vs Low Volume Accounts.


Balance Distribution: High vs Low Volume Accounts
High Volume
Low Volume
High Mean: ₹71,542
Low Mean: ₹84,088
Frequency
0
10
20
30
40
−25000
0
25000
50000
75000
100000
125000
150000
175000
Account Balance (₹)

Hypothesis Testing Based on Segmentation.

- All 3 segment pairs that are Basic, Standard and Premium are statistically different from each other.
- Premium accounts hold **₹38,074 more** than Basic on average — a massive and statistically proven gap.



Task 7: Video Presentation

- Here is the video link:
- https://www.loom.com/share/6bcf71b007a64fa99cac163febdabb86